Module 4 Assignment

1.In Module 4 structural analysis I chose the PGR and BCHE these two targets. The two proteins both from the module 2' list of candidate targets.

PGR ranks 2nd in Module 2 and has become one of the highest-ranked targets in Module 3, so I hope to further judge whether its high priority can be supported by structural evidence.

For the BCHE, it ranked 13th in the module 2, but in Module 3, it rose to 8th due to its high small molecule tractability prior (p_sm = 0.738), which is the biggest target for ranking improvement. Therefore, the purpose of this structural analysis is to further judge whether the two targets have credible small molecule binding pockets, and compare whether the structural results are consistent with the previous target prioritization.

High pLDDT and dark PAE inside the area indicate that the local structure is reliable and can be carried out for pocket analysis. If the PAE between different confident regions is very high, it means that each part itself may be correct, but the relative arrangement between them is uncertain. The area of pLDDT < 50 often does not have a stable structure, so it is not suitable for stable pocket analysis.

For PGR, AlphaFold models have 933 residues and the mean pLDDT is 56.15. Only 28.1% of the residues have a pLDDT above 90, while 63.1% of the residues are below 50. A large number of continuous low-confidence areas can be clearly seen in the pLDDT diagram, and there are also several high-confidence structural areas. Therefore, PGR cannot be regarded as a structure that is equally reliable in full length.

During the PAE and domain analysis, the jupyter notebook reports PGR has 2 confident domains, and the inter domain PAE is about 24.85Å. This value is relatively high, indicating that the relative arrangement of these structural parts is relatively uncertain. Therefore, even if the local structure of a domain itself is credible, the cross-domain pocket formed by the relative position of two domains should not be overexplained.

In contrast, BCHE's AlphaFold prediction is more reliable as a whole. BCHE has 602 residues and the mean pLDDT is 93.36, in which 83.6% residues are higher 90 and only 1.2% of residues lower than 50, and also it doesn't have an obviously long low confidence stretch. Thus, most BCHE local structures can be used in pocket analysis confidently.

BCHE notebook also reports 2 confident domains, and the inter domain PAE is about 18.55 Å. This value shows that there is still some uncertainty in the relative arrangement between different structural regions, so if a pocket must rely on the joint formation of multiple

different structural regions, I will still be cautious. However, because the pLDDT of most of the BCHE residues is very high, the pockets located within a single high-confidence area can be strongly trusted.

A large number of low pLDDT areas in PGR may be related to their regulatory and flexible regions, while higher confidence areas are more likely to correspond to more stable DNA-binding and ligand-binding domains, so pocket analysis should be mainly concentrated in these high-trust areas. BCHE is mainly composed of stable cholinesterase domains, and its overall high pLDDT is consistent with the characteristics of enzyme proteins that need to maintain a stable three-dimensional structure and catalytic function.

2. For PGR, AlphaFold prediction is compared with the experimental structure 1A28 chain A. After sequence comparison, there are 251 aligned residue pairs, of which 243 meet the consistency of amino acid and pLDDT ≥ 70 at the same time. The RMSD obtained after superposition using these confident residues is 0.95 Å. This result shows that the PGR high confidence area covered by the experimental structure is very close to the AlphaFold prediction.

Per-residue deviation shows difference between the most residues is relevantly small but there are obvious deviations in a few positions. Most difference appears in residue 864 (6.6Å), while there is also a deviation of about 2.5–5.9 Å near residues 789–793. These differences are concentrated in the part, not the whole fold. Therefore, I will not interpret these small deviations as the failure of the entire AlphaFold model, which is more likely to be local flexible conformation or differences in experimental conditions.

For BCHE, it gets 526 aligned pairs after comparing the AlphaFold structure and 6QAA chainA, and 518 of them are identical and confident residues. The whole confidence residue RMSD is 2.08Å. However, when the main high-trust area of BCHE residues 31–561 is calculated separately, RMSD is only 0.41 Å.

The largest local deviation of BCHE occurs in residue 596 (46.2 Å), while residues 403–405 also have a difference of about 2.7–3.6 Å. Because the RMSD of the core high confidence region is only 0.41 Å, the residue 596 is more like a terminal or local conformational difference than a structural error of the entire BCHE fold.

In general, the highly confident structure areas of PGR and BCHE are supported by the experimental structure. The confident RMSD of PGR is 0.95 Å, while the RMSD of the main

structural region of BCHE is 0.41 Å. Therefore, these confident regions can be used for pocket analysis.

3. Fpocket score is just the ranking of pocket, and it must also be combined with AlphaFold confidence. The pocket with mean pLDDT <70 were rejected and PAE within > 10 Å suspect and other keep.

PGR's Pocket 1 druggability score is 0.834, mean pLDDT is 95.68, and PAE within is 1.70 Å, so it is judged to be keep. In contrast, although Pocket 16 entered the top five, the mean pLDDT was only 69.70, so it was rejected. This shows that confidence filtering excludes structurally unreliable pockets.

BCHE's Pocket 1 score is 0.956, mean pLDDT is 96.68, and PAE within is 1.21 Å, so it is also judged to be keep. The top five pockets of BCHE all pass the confidence filter.

Therefore, structurally, this target has a reliable small molecule binding site, so it has the potential to be targeted by small-molecule drugs.

PGR: top five pockets by druggability score

pocket	Druggability Score	Volume	mean_pLDDT	PAE_within	verdict	rank
1	0.834	554.086	95.684545	1.704545	keep	1
17	0.009	513.316	83.645000	9.166667	keep	2
4	0.006	505.692	94.597500	1.808594	keep	3
16	0.006	273.510	69.703000	6.670000	reject	4
19	0.006	421.734	94.446667	1.854167	keep	5

BCHE: top five pockets by druggability score

pocket	Druggability Score	Volume	mean_pLDDT	PAE_within	verdict	rank
1	0.956	790.404	96.681818	1.206612	keep	1
2	0.587	833.019	97.898889	1.532236	keep	2
29	0.054	873.234	84.448824	8.647059	keep	3
25	0.051	238.784	72.672500	1.625000	keep	4
19	0.025	333.868	72.473333	6.250000	keep	5

4. PGR already has a high ranking in Module 2, and Module 4 further finds that its high-trust structure area matches the experimental structure well, and there is a reliable high-scoring pocket. Therefore, the structural results support the continued retention of PGR.

BCHE only ranked 13th in Module 2, but rose to 8th in Module 3 due to its high small-molecule prior (p_sm = 0.738). Module 4 shows that the overall structure of BCHE is very reliable, the core structure is highly consistent with the experimental results, and there are multiple reliable pockets. Therefore, the structural evidence of Module 4 supports Module 3's reranking of BCHE.

Overall, I will continue to keep the two targets of PGR and BCHE.

5. The first limitation is the one snapshot mentioned in Lecture slide 22. AlphaFold only provides a static configuration, but the real protein will change dynamically. This problem is especially important for PGR because it has a large number of low pLDDT regions and high inter-domain PAE. In the future, we can use molecular dynamics or compare multiple experimental structures to study different configurations.

The second limitation is missing ligands or partners. AlphaFold prediction usually does not include ligand, metal or other binding partners in the experimental structure, so local structural differences do not necessarily represent prediction errors. It can be further verified by comparing apo and ligand-bound structures, or by molecular dynamics and ligand-binding experiments.

6. For PGR, the ligand site in the experimental structure is completely covered by Pocket 1, and the coverage is 1.00. Pocket 1 is also rank 1, the druggability score is 0.834, and the verdict is keep. Therefore, PGR's best kept pocket is completely consistent with the experimental ligand-binding site, which provides direct support for its small-molecule druggability.

The results of BCHE are different. The highest score of Pocket 1 (score = 0.956) did not cover the experimental HUN A613 ligand site. What really completely covers the HUN site is Pocket 2, with rank 2, score 0.587, verdict keep, and coverage 1.00.

This means that the highest-ranked pocket of fpocket is not necessarily the real functional binding site. Real pocket may get a lower geometric score because it is more open, shallower or more polar.

p53 Anchor Check：

The RMSD of the AlphaFold model and experimental structure 1TUP chain B of p53 on confident residues is 0.68 Å, indicating the prediction of DNA-binding domain. It is very close to the experimental structure. Among the hot residues, C176, H179, C238 and C242 are zinc ligands, while R248 and R273 are in direct contact with DNA, and R249 is located near DNA. These residues are located in the high confidence area of AlphaFold, indicating that they have an important structural or functional role, and also help explain why some hotspot variants in Module 3 have strong sequence-based effects. However, these variant scores mainly reflect sequence tolerance and are not directly equivalent to clinical pathogenicity.